Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Development of Scanning-Free Stagnation Flow Spray Coater for Photoresist or Adhesive in a Semi-conductor Process**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1 (**Large Category**)**		**Technology Field 2 (**Large Category**)**	
	Machinery and materials			
	Medium Category	Small Category	Medium Category	Small Category
	Semiconductor Equipment	***Deposition Equipment***		
Concept of Proposed Technology	**Core Concept** While traditional spray coating offers high throughput, its adoption in advanced packaging is limited by the poor uniformity inherent to raster-scanning. We propose a scanning-free spray coater utilizing a 1D stagnation flow configuration coupled with a rotating disk. By delivering micro-droplets (< 5 m) within a CVD-inspired fluid dynamic regime, this technology achieves a theoretically uniform deposition flux across the entire substrate. **Key Functions and Potential Applications** If successful, this method will deliver high-yield uniformity (< 2 %) comparable to spin coating for: ● Advanced Packaging (HBM): Precise, uniform adhesive and dielectric layers for high-density			

	stacking. ● MEMS Fabrication: Conformal coverage of high-aspect-ratio features and complex 3D topographies. ● Large-Format Substrates: Uniform photoresist application on large-diameter or non-circular wafers where spin coating is unviable.
Rationale for Joint R&D Project	**Critical Technical Barriers** While spin coating remains the industry standard for planar photoresist deposition, it faces fundamental limitations in high-aspect-ratio 3D architectures. In contrast, spray coating offers transformative advantages for 2.5D/3D and Fan-Out Wafer-Level Packaging (FOWLP), including superior conformal coverage over complex topography, significantly enhanced material utilization, and improved thickness consistency at the wafer periphery. However, current ultrasonic spraying is primarily restricted to non-critical layers; it lacks the global surface uniformity and 'mirror-like' finish required for high-precision sub-micron lithography. Unlike spin-coating, which utilizes continuous centrifugal flow to achieve sub-percent global uniformity, spray coating is a discontinuous deposition process. The primary challenge lies in the stochastic coalescence of atomized droplets, where millions of discrete liquid particles must merge into smooth film. Inherent variabilities in droplet size distribution, evaporation rates during flight, and nozzle scanning overlap frequently result in localized thickness variations, typically ranging from 5 to 10 % which can compromise downstream high-resolution lithography. The non-uniformity of the spray could be categorized as: ● "Orange Peel" Phenomenon (Surface Roughness): If droplets dry prematurely before they can fully "wet" and level on the substrate, the film develops a granular texture. This surface morphology prevents uniform contact with photomasks and causes light scattering during exposure, leading to critical dimension (CD) errors. ● Scanning/Raster Striping: Because the nozzle moves in a programmed path (zigzag or spiral), any minor fluctuation in pump pressure or travel speed creates linear thickness bands. This "striping" effect creates a periodic non-uniformity that is difficult to compensate for in subsequent etching steps. ● The "Gaussian" Cone Profile of the spray: Most spray nozzles produce a Gaussian distribution of material (thicker in the center, thinner at the edges). Achieving a flat profile requires complex mathematical "stitching" of overlapping spray passes, which is highly sensitive to the nozzle-to-substrate distance. The key advantages of spray coating are as follows: ● Conformal Coverage for 3D Topography Spin coating often suffers from "shadowing" or pooling in high-aspect-ratio features. In contrast, ultrasonic or pneumatic spray coating provides superior conformal coverage over complex 3D microstructures, such as Through-Silicon Vias (TSVs) and deep trenches, ensuring consistent resist thickness regardless of surface geometry. ● Substantial Material Cost Reduction Spin coating is inherently wasteful, with up to 90% of the photoresist lost to "edge fling." Spray coating is a direct-deposition method that achieves 60%

	to 80% material savings. This drastically reduces the Cost of Ownership (CoO), especially when using expensive specialized resists. ● Elimination of Edge Bead & Turbulence Effects Spray coating eliminates the "edge bead" effect, removing the need for costly edge-bead removal (EBR) steps and improving yield at the wafer periphery. ● Scalability For large-format wafers (300mm/12-inch and above), spin coating creates air turbulence at high RPMs, leading to thickness variations. Spray coating remains independent of rotational physics, maintaining uniformity across the entire diameter. ● Geometric Versatility Unlike spin coating, which is restricted to circular substrates to avoid centrifugal vibration and "corner buildup," spray coating is compatible with rectangular substrates and panel-level packaging (PLP). The primary barriers to spray coating adoption are as follows: ● The uniformity gap. Achieving thin coating (0.1–10㎛) with excellent uniformity of spin coating is physically challenging due to the thickness variations inherent in multi-pass scanning overlaps. ● It is challenging for design of the stagnant flow sprayer system with assistance of Computational Fluid Dynamics (CFD) simulation and the characterization of its performance, specifically focusing on velocity fields, spray pattern, and droplet size distribution. The spray characterization instruments such as PDPA is needed for velocity filed and drop size. ● It is challenging to integrate the stagnant flow sprayer system to spin coaters. This includes developing a control system for fluid flow and temperature, as well as a mechanical system for exhaust and rotation. **Why Global Collaboration?** ● Global leading institute is needed to lead the coating performance testing, evaluating film thickness and uniformity while identifying potential technical challenges.
Key Words	1. Photoresist 2. Advanced Packaging 3. Spray Coating 4. Stagnant Flow 5. Scanning Free

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]	**Start of the Project**	Level (3)	**End of the Project**	Level (5)

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R&D Objectives	**R&D Objectives** We propose the development of a scanning-free spray coating system inspired by rotating disk Chemical Vapor Deposition (CVD) reactors. By leveraging 1D laminar stagnation flow dynamics, this device achieves theoretically perfect deposition uniformity on a rotating substrate. Utilizing an ultrasound sprayer to generate a fine mist (1–5 μm), the droplet mixture mimics gas-phase behavior at operational velocities (Stokes No. St << 1). We will employ Computational Fluid Dynamics (CFD) simulations to optimize the stagnation flow geometry, ensuring a scalable, high-uniformity coating solution. **Detailed Objectives** The implementation of a scanning-free, stagnation-flow spray coater represents a paradigm shift in thin-film deposition: ● Superior Uniformity in Advanced Packaging: Unlike conventional scanning sprayers that suffer from "overlap" inconsistencies, the stagnation flow model provides radially symmetric uniformity. This is critical for 2.5D/3D packaging and Fan-Out Wafer-Level Packaging (FOWLP) where topography is complex. ● High-Viscosity Capability: Standard spin-coating often struggles with high-viscosity photoresists or dielectric materials, leading to excessive waste and "edge bead" issues. This technology enables the deposition of thick, viscous films with minimal material loss. ● Increased Throughput: By eliminating the need for a mechanical scanning arm (the X-Y motion), the process significantly reduces cycle times and mechanical points of failure, directly improving Cost of Ownership for IC fabrication. ● Versatility across Substrates: This method can provide uniform coverage on non-circular substrates or textured surfaces (like solar cells) that are traditionally difficult for spin-coating to handle.
R&D Target Specifications	The project aims to achieve the following four technical milestones: ● Fine Atomization: Engineer a spray system capable of producing droplet sizes between 1–5 μm to ensure gas like a behavior of the droplet. Testing fluid will be a photoresist (AZ 1518-MicroChemicals) and thermoplastic resin (WaferBOND, HT-10.10, Brewer Science). ● Uniformity of Coating Thickness: The primary technical milestone for the prototype will be achieving a local uniformity with 1 %, and Total Thickness Variation below 2 % across a 200 mm substrate, a 5 times improvement over industry-standard raster-spray systems.
R&D Plan	● Proposed Technology: 1D Stagnation Flow Spray Coater with Rotating Disk To overcome the inherent non-uniformity and material waste of conventional raster-scanning spray systems, we propose a Scanning-Free 1D Stagnation Flow Configuration. This design utilizes a stationary showerhead inlet and a rotating substrate disk to create a "uniform-flux" deposition environment. ■ Fluid Dynamic Foundation The proposed system is based on the Von Kármán swirling flow theory. In this

configuration: The rotating disk acts as a centrifugal pump, pulling the spray mist axially toward the surface and pushing it out radially. When the incoming spray mist is treated as a continuous medium, the boundary layer thickness (δ) becomes independent of the radial location

$$\delta = \sqrt{\frac{\nu}{\omega}}$$

where ν is kinematic viscosity and ω is angular velocity.

Because the boundary layer is uniform across the entire disk, the deposition rate remains constant from the center to the edge, achieving "perfect" theoretical uniformity without the need for nozzle movement.

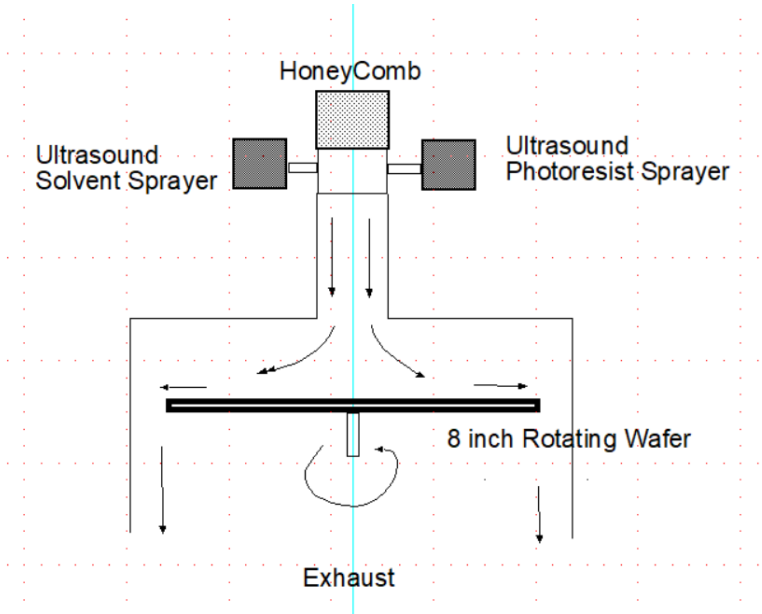

Figure 1. Schematic of the 1D stagnation flow sprayer.

The proposed Stagnation Flow Spray Coater draws inspiration from the principles of Stagnation Flow Chemical Vapor Deposition (CVD) reactors, established by Chown et al. (1986) and Houtman et al. (1988). By leveraging Computational Fluid Dynamics (CFD) modeling, Luo et al. (2004) demonstrated that optimizing stagnation flow geometry and rotation speed can reduce surface roughness (RMS) variation to as low as 1% under atmospheric pressure and heated substrate conditions. Fig. 2 illustrates a sample geometry optimized for such uniform deposition.

The primary challenge in adapting this CVD-inspired model to a spray system is the physical distinction between liquid spray droplets and gas-phase species. We hypothesize that below a critical particle size for a given velocity field, floated droplets will exhibit aerodynamic behavior analogous to gas species (Stokes No, St<<1), adhering to the established stagnation flow models. The validity of this gas-behavior assumption will be a primary focus of the performance validation phase. We will experimentally determine the droplet size thresholds required to maintain the 1% uniformity predicted by the CFD models, ensuring that the advantages of spray scalability are achieved without sacrificing

the precision of traditional spin-coating or CVD methods.

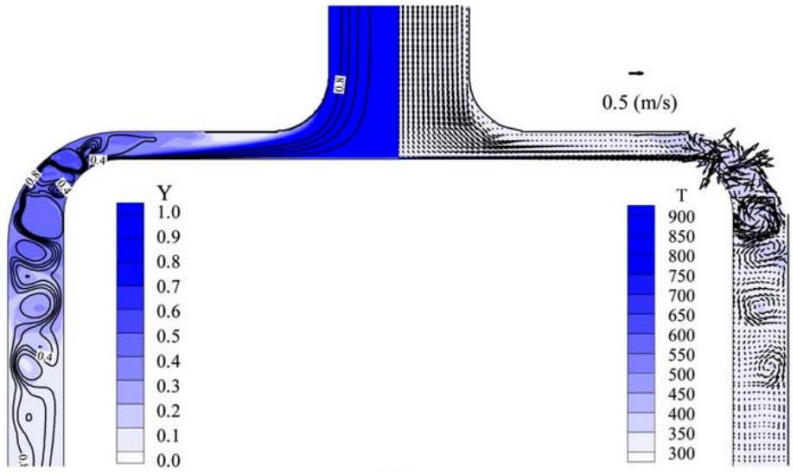

Figure 2. A Sample optimized geometry for the atmospheric CVD reactor.

■ Mitigation of 'Orange Peel' and Raster Defects

Traditional sprayers suffer from "striping" due to the Gaussian nature of the spray cone and the overlap of scanning paths. Our design eliminates these by:

- Scanning-Free Operation: The stationary showerhead eliminates mechanical jitter and overlap inconsistencies.
- In-Situ Solvent Control: By introducing a controlled ratio of solvent spray mist alongside the photoresist mist, we maintain a "wet" surface environment. This prevents the premature drying of droplets—the primary cause of orange peel—allowing the liquid to level out via surface tension before solidification.
- Rotational Smoothing: The low-speed rotation (lower than standard spin-coating, < 1000 rpm) provides just enough centrifugal force to assist in surface leveling without "flinging" the majority of the material off the wafer.

■ Design Optimization via CFD

The transition from a 3D spray environment to a 1D stagnation field requires precise geometric tuning. We will utilize Computational Fluid Dynamics (CFD) to optimize:

- H/D Ratio: The distance between the showerhead (H) and the disk (D) to ensure the flow reaches a steady stagnation state before impact.
- Honeycomb Integration: Designing the flow-straightening honeycomb to eliminate turbulence and ensure a "plug flow" velocity profile.
- Flow Balancing: Matching the inlet velocity of the mist with the rotation-induced suction of the disk to prevent recirculation zones or buoyancy-driven instabilities.

● CFD simulation for geometry design

The rotating geometry will be simulated using Ansys Fluent, utilizing a comprehensive fluid model that incorporates spray droplet dynamics and coalescence effects. The CFD model will be validated against a physical prototype by measuring the velocity field and droplet size

distribution near the surface of the rotating disk.

This simulation will provide critical insights into the interplay between inlet velocity profiles, housing geometry, and rotation speeds. Furthermore, the model will quantify the impact of varied particle sizes and distributions on the final film uniformity, enabling the precise optimization required to achieve spin-coater-level performance.

- Fabrication and Design Optimization

  CFD simulations will guide the initial geometric design for 6-inch wafer, while final optimization will be achieved through experimental tuning. Spray diagnostics instrument will be used to validate the CFD result. Coating thickness measurements will be used to calibrate the solvent-to-photoresist ratio and rotation speed. Additionally, droplet size will be optimized by adjusting the ultrasonic sprayer frequency to ensure precise deposition.

- Performance Validation

  Purdue University is home to the Birck Nanotechnology Center (BNC), which houses the Scifres Nanofabrication Laboratory. This 25,000-square-foot facility is one of the largest and most advanced academic cleanrooms in the United States, featuring ISO Class 3 (Class 1) bays.

  Optical Profilometry: To prove the proposed "1D Stagnation Flow" sprayer, Optical Profilometry will be used such as Bruker GT-K and KLA-Tencor P-10 for measuring global TTV (Total Thickness Variation) across the wafer.

  Scanning Electron Microscopy (SEM/TEM): For cross-sectional analysis of how well your spray mist coats the sidewalls of deep trenches or TSVs.

  The performance of spray coater will be tested using a photoresist (AZ 1518-MicroChemicals) and thermoplastic resin (Wafer BOND, HT-10.10, Brewer Science).

- System Integration

  A prototype system that is optimized through validation will be integrated one of the existing model that SRC-3000 or SRC-4000 to demonstrate the system to their customer. The existing system will be modified to house spray control system, and hood and exhaust system to demonstrate to the customer.

- Milestone

Task/ Months from Start	6	12	18	24	30	36
CFD Analysis/Design						
Fabrication/Optimization						
Performance Validation						
System Integration						

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	Research Institute will provide base design and fabrication for the stagnation flow sprayer with the help of CFD simulation. Few ultrasound sprayers will be purchased and evaluated using fluid and characterize the drop size, velocity of the droplet and spatial distribution of the spray droplet. In addition, Research Institute will provide controller for flow, pressure and temperature, and temporary coating stand to spray onto a wafer. Purdue University will evaluate the performance of a spray coater by applying thermoplastic adhesives and photoresist onto various wafer sizes. Following the spray process, the coated wafers will be baked. Success will be determined by measuring coating thickness and uniformity using a surface profiler.
Expected R&R of Korean SME Partner	Korean SME Partner will integrate the spray dispenser into a complete system, developing custom controls for flow, pressure, and temperature. The integration will include a rotating mechanism and an exhaust control system, all managed by a dedicated, custom-developed control software.

5. Expected Outcomes/Impact

Expected Outcomes and Impact

The market for a high-uniformity spray coating system is poised for significant growth as semiconductor manufacturing shifts toward 3D architectures, flexible substrates, and sustainable processes. While spin coating remains the industry standard for flat wafers, it is physically limited when dealing with the complex geometries required for modern electronics.

- Market Opportunity & Drivers
  - 3D Structures & MEMS: As devices move toward Heterogeneous Integration and Advanced Packaging, surfaces are no longer flat. Spin coating creates "shadowing" and uneven thickness on sidewalls or deep trenches. A high-uniformity spray system provides conformal coverage that spin coating cannot match.
  - Flexible Electronics: The rise of wearables and foldable displays requires coating on flexible materials. Spin coating is incompatible with Roll-to-Roll (R2R) processing, whereas spray coating is the primary solution for continuous manufacturing.
  - Material Waste Reduction: Spin coating is notoriously inefficient, often wasting 90% to 95% of expensive photoresists. The proposed system can reduce material consumption by over 50%, offering a massive ROI for high-volume manufacturers

Segment	Why they need this	Estimated Impact
Semiconductors (MEMS & 3D)	Spin coating fails on 3D structures (cavities/vias). Your tech allows uniform photoresist on stepped wafers.	High Value: Precision is non-negotiable here.
Flexible Electronics	Roll-to-roll (R2R) manufacturing requires spray/slot-die, but uniformity issues often lower device yield.	High Growth: Essential for foldable displays and wearables.

- Market Size & Economic Drivers

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- Spin Coating Market: Valued at approximately $300M - $350M (2025/2026). This is the "displacement" target. If the proposed spray coater is as uniform as a spin coater, it can capture the portion of this market that is frustrated by 90% material waste (standard in spin coating).
- Key Economic Advantage:
  - Material Savings: Spin coating wastes the vast majority of the fluid. In industries using expensive "inks" (like gold nanoparticles or advanced polymers), a uniform spray coater could reduce material costs by 50–80%.
- Key Competitive Advantages
  - To capture this market, the proposed sprayer value proposition should focus on these two pillars:
  - Topography Handling: Unlike spin coaters, you can coat 3D surfaces, curved glass, and trenches with the same precision.
  - Yield Improvement: By matching spin-coating uniformity (1-2 % variation), you eliminate the "mottle" or "orange peel" effects that typically cause device failure in standard spray systems.

6. References

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